

v3 post-BDT Armenteros study

K_S^0 pair contamination in 1,028 $Z \rightarrow b\bar{b}$ chunks

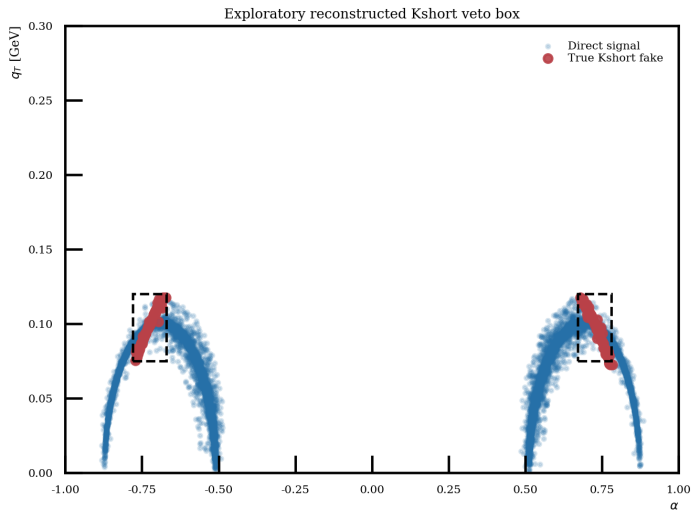
FCC-ee $\Lambda_b \rightarrow \Lambda\gamma$ analysis

3 October 2026

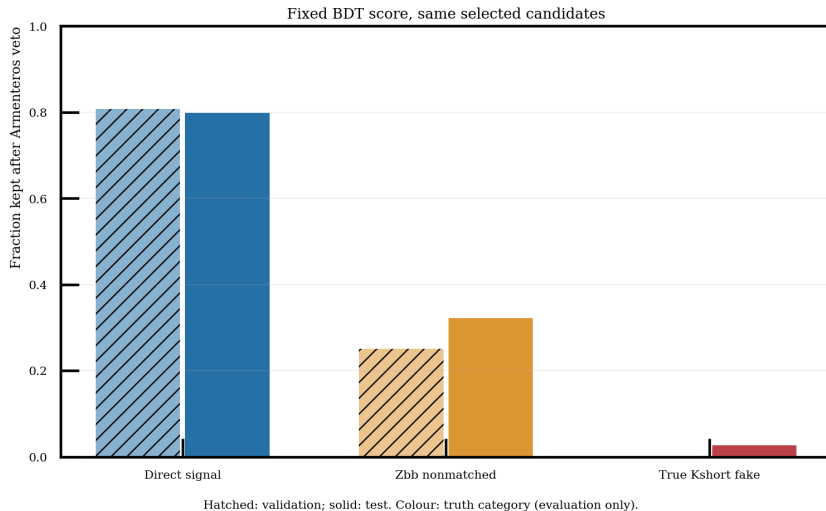
Fixed input and question

- Offline: $|m(p\pi) - m_\Lambda| \leq 12.5$ MeV, $E_{\Lambda_b} \geq 10.5$ GeV; no photon veto.
- Frozen XGBoost score ≥ 0.9787055254 ; 100,000 forced signal events and 376,723,929 processed Zbb input events.
- After score: 228 nonmatched Zbb candidates, including 165 true $K_S^0 \rightarrow \pi\pi$ pairs assigned $p\pi$.
- Veto proposal: $0.67 \leq |\alpha| \leq 0.78$ and $0.075 \leq q_T \leq 0.120$ GeV. Reconstructed fitted-track variables only; ancestry is diagnostic.

Armenteros geometry after the fixed score



Validation and test retention



Counted effect and PI decision

Test partition	Direct signal	Other Zbb	True K_S^0 pair
Fixed score	5,719	56	37
Also Armenteros veto	4,568	18	1

- Projected test S/B: 233,460/657,999 before; 186,474/211,500 after. Central purity 26.2% $\rightarrow$ 46.9%.
- The 18 test Zbb candidates imply a 95% Poisson B interval 125,348–334,261. Z_{cc}/Z_{ss} are absent.
- About 20% of direct signal is lost. The box was explored after inspecting both partitions; the test is descriptive, not a blinded veto validation.
- Request: inspect this tradeoff, compare the reconstructed $\pi\pi$ mass veto, then validate on fresh chunks before adoption.